

Text Representation Error Analysis

- **Main failure pattern**

Both Sentence-BERT and BioBERT struggled with the minority "maybe" class.

- **BioBERT**

BioBERT validation macro-F1 was 0.426 and test macro-F1 was 0.375.

The confusion matrix shows frequent confusion between "yes" and "no", and limited correct prediction of "maybe".

- **Sentence-BERT**

Sentence-BERT validation macro-F1 was 0.394 and test macro-F1 was 0.384.

It predicted more "maybe" cases than BioBERT on the test set, but still confused many uncertain cases with yes/no labels.

- **Interpretation**

Dense representations improved semantic modelling compared with sparse baselines, but embedding quality alone was not enough. PubMedQA requires deeper question-context interaction and statistical reasoning, especially for ambiguous medical evidence.